数列极限的 Stolz 公式

定理 1 ($\frac{0}{0}$**型)**设 $\{x_n\}$ 严格单减，且 $\lim\limits_{n\to\infty} x_n = 0$，$\lim\limits_{n\to\infty} y_n = 0$. 若

$$\lim_{n\to\infty} \frac{y_n - y_{n-1}}{x_n - x_{n-1}} = a ,$$

则 $\lim\limits_{n\to\infty} \frac{y_n}{x_n} = a$.

定理 2 ($\frac{\infty}{\infty}$**型)**设 $\{x_n\}$ 严格单增，且 $\lim\limits_{n\to\infty} x_n = +\infty$. 若

$$\lim_{n\to\infty} \frac{y_n - y_{n-1}}{x_n - x_{n-1}} = a ,$$

则 $\lim\limits_{n\to\infty} \frac{y_n}{x_n} = a$.

证明见数学分析类教材.

应用举例

例 1 若 $\lim\limits_{n\to\infty} a_n = a$，则 $\lim\limits_{n\to\infty} \frac{a_1 + a_2 + \cdots + a_n}{n} = a$.

事实上令 $x_n = n, y_n = a_1 + a_2 + \cdots + a_n$，则 $\lim\limits_{n\to\infty} x_n = +\infty$.因为

$$\lim_{n\to\infty} \frac{y_n - y_{n-1}}{x_n - x_{n-1}} = \lim_{n\to\infty} \frac{a_n}{n-(n-1)} = \lim_{n\to\infty} a_n = a ,$$

所以 $\lim\limits_{n\to\infty} \frac{y_n}{x_n} = a$，即 $\lim\limits_{n\to\infty} \frac{a_1 + a_2 + \cdots + a_n}{n} = a$.

例 2 求极限 $\lim\limits_{n\to\infty} \frac{1^m + 2^m + \cdots + n^m}{n^{m+1}}$（$m$ 为正整数）.

解 令 $x_n = n^{m+1}, y_n = 1^m + 2^m + \cdots + n^m$，则 $\lim\limits_{n\to\infty} x_n = +\infty$.因为

$$\lim_{n\to\infty} \frac{y_{n+1} - y_n}{x_{n+1} - x_n} = \lim_{n\to\infty} \frac{(n+1)^m}{(n+1)^{m+1} - n^{m+1}} = \lim_{n\to\infty} \frac{n^m + C_m^1 n^{m-1} + \cdots + 1}{(n^{m+1} + C_{m+1}^1 n^m + \cdots + 1) - n^{m+1}}$$

$$= \lim_{n\to\infty} \frac{n^m + C_m^1 n^{m-1} + \cdots + 1}{C_{m+1}^1 n^m + C_{m+1}^2 n^{m-1} \cdots + 1}$$

$$= \lim_{n\to\infty} \frac{1 + C_m^1 \cdot \frac{1}{n} + \cdots + \frac{1}{n^m}}{(m+1) + C_{m+1}^2 \cdot \frac{1}{n} + \cdots + \frac{1}{n^m}} = \frac{1}{m+1}.$$

例 3 求极限 $\lim_{n\to\infty}(\frac{1}{n+1^\alpha} + \frac{1}{n+2^\alpha} + \cdots + \frac{1}{n+n^\alpha})$ $(\alpha > 1)$.

解 记数列为 z_n. 注意 $n + k^\alpha \geq 2\sqrt{n \cdot k^\alpha} = 2k^{\alpha/2}\sqrt{n}$ 有

$$z_n \leq \sum_{k=1}^{n} \frac{1}{2k^{\alpha/2}\sqrt{n}} = \frac{1}{2} \frac{\sum_{k=1}^{n} \frac{1}{k^{\alpha/2}}}{\sqrt{n}}.$$

而应用 Stolz 公式有

$$\lim_{n\to\infty} \frac{\sum_{k=1}^{n} \frac{1}{k^{\alpha/2}}}{\sqrt{n}} = \lim_{n\to\infty} \frac{\sum_{k=1}^{n+1} \frac{1}{k^{\alpha/2}} - \sum_{k=1}^{n} \frac{1}{k^{\alpha/2}}}{\sqrt{n+1} - \sqrt{n}} = \lim_{n\to\infty} \frac{\frac{1}{(n+1)^{\alpha/2}}}{\sqrt{n+1} - \sqrt{n}}$$

（有理化为）$= \lim_{n\to\infty} \frac{\sqrt{n+1} + \sqrt{n}}{(n+1)^{\alpha/2}} = 0$ (注意 $\frac{\sqrt{n+1} + \sqrt{n}}{(n+1)^{\alpha/2}} = \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1}} \cdot \frac{1}{(n+1)^{\frac{\alpha-1}{2}}}$)

由夹挤准则知 $\lim_{n\to\infty} z_n = 0$.